FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 18

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Recap *of previous lecture*

INDEFINITE INTEGRATION



Topics to be covered

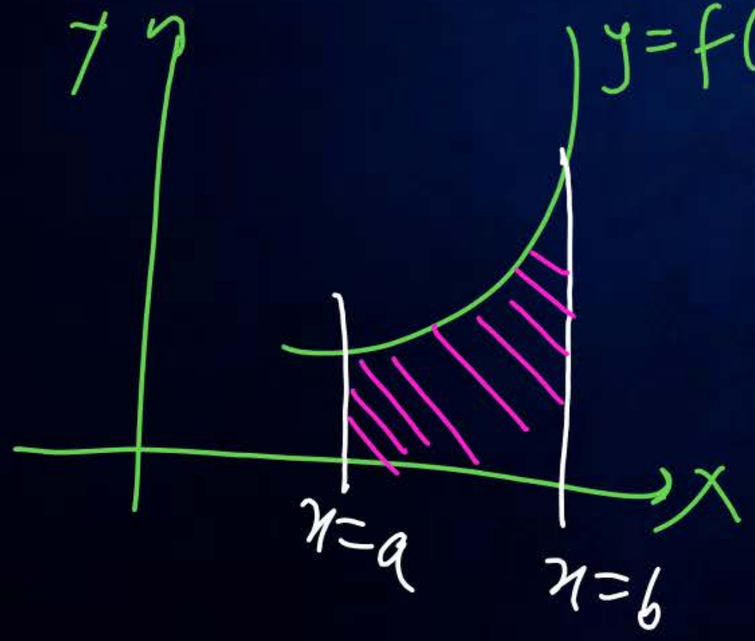
"DEFINITE INTEGRATION"



Fundamental Theorem of Integral Calculus →

$$\int_a^b f(x)\,dx = \Big[\phi(x)+c\Big]_{x=a}^{b} = \{\phi(b)+c\} - \{\phi(a)+c\} = \phi(b)-\phi(a)$$

i.e. $\boxed{\int_a^b f(x)\,dx = \phi(b)-\phi(a)}$ = Area under $f(x)$ b/n $x=a$ and $x=b$



Properties of Definite Integration →

① $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$

Put $x=t$
$dx=dt$
At $x=a$, $t=a$
at $x=b$, $t=b$

② $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

③ $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

where $a < c < b$

④ $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

⑤ $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

⑥ $\int_0^a x f(x)\,dx = \frac{a}{2}\int_0^a f(x)\,dx$

where $f(a-x) = f(x)$

Proof (4) Put $a+b-x = t$ ⟶ at $x=a$, $t=b$; at $x=b$, $t=a$

$-dx = dt$

$RHS = \int_a^b f(a+b-x)\,dx = \int_b^a f(t)(-dt) = \int_a^b f(t)\,dt$

$= LHS.$

Proof ⑥ $I = \int_0^a x f(x)\,dx = \int_0^a (a-x) f(a-x)\,dx$

$I = \int_0^a a f(a-x)\,dx - \int_0^a x f(a-x)\,dx$

$I = a\int_0^a f(x)\,dx - \int_0^a x f(x)\,dx$

$2I = a\int_0^a f(x)\,dx \Rightarrow I = \frac{a}{2}\int_0^a f(x)\,dx$

(1) $\int_0^{2a} f(x)\,dx = \int_0^{a} [f(x)+f(2a-x)]\,dx = \begin{cases} 2\int_0^a f(x)\,dx, & \text{if } f(2a-x)=f(x) \\ 0, & \text{if } f(2a-x)=-f(x) \end{cases}$

Learn.

P.T $\int_a^{2a} f(x)\,dx = \int_0^a f(2a-x)\,dx$

Let $2a-x=t$ → at $x=0$, $t=2a$; at $x=a$, $t=a$

$-dx=+dt$

So RHS $= \int_0^a f(2a-x)\,dx = \int_{2a}^{2a} f(t)(-dt)$

$= \int_a^{2a} f(t)\,dt = \int_a^{2a} f(x)\,dx$

i.e. $\int_0^{2a} f(x)\,dx = \int_0^a f(x)\,dx + \int_a^{2a} f(x)\,dx = \int_0^a f(x)\,dx + \int_0^a f(2a-x)\,dx = \int_0^a [f(x)+f(2a-x)]\,dx$

(2) $\int_{-a}^{a} f(x)\,dx = \int_0^a [f(x)+f(-x)]\,dx$

$= \begin{cases} 2\int_0^a f(x)\,dx, & \text{if } f(-x)=+f(x) \\ 0, & \text{if } f(-x)=-f(x) \end{cases}$

if $f(-x)=f(x)$ then $f(x)$ is said to be an Even func^n

if $f(-x)=-f(x)$ " " " " " odd func^n

General Question :-

Q1 $I = \int_0^{\pi/4} \tan^2 x\,dx = ? = \int_0^{\pi/4} (\sec^2 x - 1)\,dx$

$= (\tan x - x)_0^{\pi/4}$

$= \left(\tan\frac{\pi}{4} - \frac{\pi}{4}\right) - (\tan 0 - 0)$

$= 1 - \frac{\pi}{4}$

Q2 $I = \int_0^{\pi/4} \sqrt{1+\sin 2x}\,dx = ?$

Sol: W.K.T;

$1 + \sin 2x = (\cos^2 x + \sin^2 x) + 2\sin x \cos x$

$= (\cos x + \sin x)^2$

$\sqrt{1+\sin 2x} = \cos x + \sin x$

Hence $I = \int_0^{\pi/4} (\cos x + \sin x)\,dx$

$= (\sin x - \cos x)_0^{\pi/4}$

$= \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) - (0 - 1) = 1$

Q. $I = \int_0^{\pi/2} \sqrt{\cos\theta}\,\sin^3\theta\,d\theta = ?$

Put $\cos\theta = t$

At $\theta = 0$, $t = 1$

At $\theta = \frac{\pi}{2}$, $t = 0$

$-\sin\theta\,d\theta = dt$

$$I = \int_0^{\pi/2} \sqrt{\cos\theta}\cdot\sin^2\theta\cdot\sin\theta\,d\theta$$

$$= \int_1^0 \sqrt{t}\cdot(1-t^2)(-dt)$$

$$= -\int_0^1 \left(t^{\frac{1}{2}} - t^{\frac{5}{2}}\right)(-dt)$$

$$= \int_0^1 \left(t^{\frac{1}{2}} - t^{\frac{5}{2}}\right)dt$$

$$= \left(\frac{t^{\frac{3}{2}}}{\frac{3}{2}} - \frac{t^{\frac{7}{2}}}{\frac{7}{2}}\right)_0^1$$

$$\left(\int t^a\,dt = \frac{t^{a+1}}{a+1}\right)$$

$$= \left\{\frac{2}{3}(1) - \frac{2}{7}(1)\right\} - \{0-0\}$$

$$= 2\left(\frac{7-3}{21}\right) = \frac{8}{21}$$

Prop 1 & 2 →

eg $I = \int_1^4 (x)\,dx = \left(\frac{x^2}{2}\right)_1^4 = \left(\frac{16}{2} - \frac{1}{2}\right) = \frac{15}{2}$

$I = \int_1^4 (t)\,dt = \left(\frac{t^2}{2}\right)_1^4 = \left(\frac{16}{2} - \frac{1}{2}\right) = \frac{15}{2}$

$I = \int_0^{\pi/2} (\sin x)\,dx = (-\cos x)_0^{\pi/2}$

$= -\{0 - 1\} = 1$

$I = \int_0^{\pi/2} \sin\theta\,d\theta = \cdots = 1$

Prop 3 →

$I = \int_0^4 x\,dx = ?$

(M-I) $I = \int_0^4 x\,dx = \left(\frac{x^2}{2}\right)_0^4 = \frac{16}{2} - 0 = 8$

(M-II) $I = \int_0^4 x\,dx = \int_0^2 x\,dx + \int_2^4 x\,dx$

$= \left(\frac{x^2}{2}\right)_0^2 + \left(\frac{x^2}{2}\right)_2^4$

$= (2 - 0) + (8 - 2) = 8$

Hence Verified.

Q. $I = \int_0^{\pi} |\cos x|\, dx = ?$

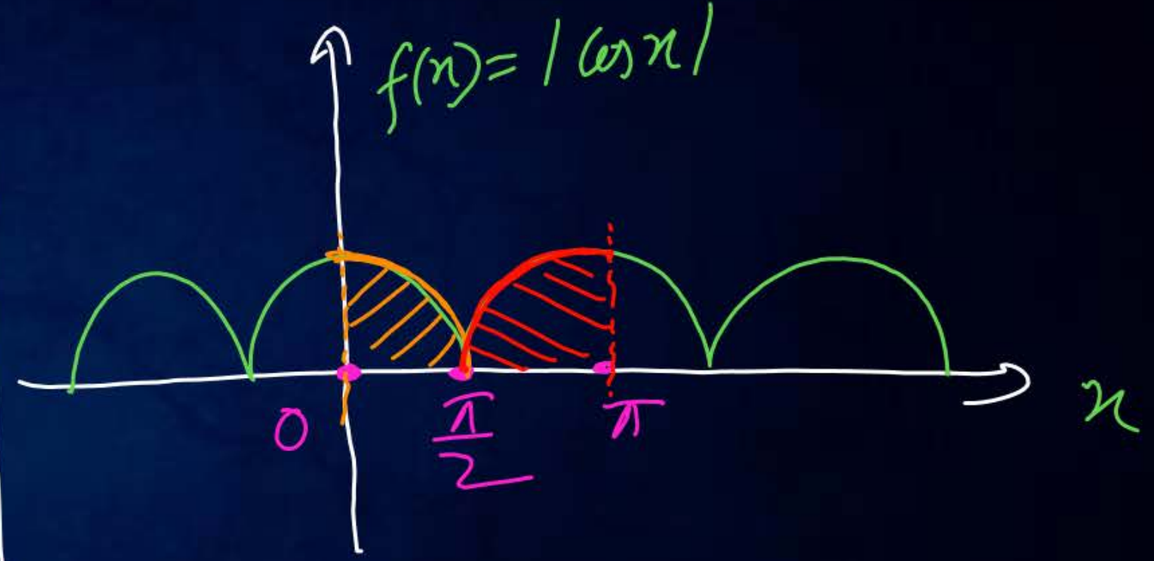
$\because |\cos x| = 0$ at $x = \frac{\pi}{2}$ also,

ie $|\cos x| = \begin{cases} +\cos x, & 0 < x < \frac{\pi}{2} \\ -\cos x, & \frac{\pi}{2} < x < \pi \end{cases}$

$$I = \int_0^{\pi} |\cos x|\, dx = \int_0^{\pi/2} (\cos x)\, dx + \int_{\pi/2}^{\pi} (-\cos x)\, dx$$

$$= (\sin x)_0^{\pi/2} - (\sin x)_{\pi/2}^{\pi}$$

$$= (1-0) - (0-1) = 2$$



Qe $I = \int_{-1}^{2} (1+|x|)\,dx = ?$

$\because |x| = \begin{cases} -x, & x < 0 \\ +x, & x \geq 0 \end{cases}$

$I = \int_{-1}^{0} (1-x)\,dx + \int_{0}^{2} (1+x)\,dx$

$= \left(x - \frac{x^2}{2}\right)_{-1}^{0} + \left(x + \frac{x^2}{2}\right)_{0}^{2}$

$= \left[\left(0 - \frac{0}{2}\right) - \left(-1 - \frac{1}{2}\right)\right]$

$= \left[(2+2) - (0+0)\right]$

$= \frac{3}{2} + 4 = \frac{11}{2} = 5.5$

Quest Based on Prop 4 & 5 ⇒

Qe $I = \int_{0}^{\pi/2} \left(\frac{\sin x}{\sin x + \cos x}\right) dx = ?$ — (1)

By Prop (5), $x \to \left(\frac{\pi}{2} - x\right)$

$I = \int_{0}^{\pi/2} \frac{\cos x}{\cos x + \sin x}\,dx$ — (2)

$I = \frac{\pi}{4}$

$2I = \int_{0}^{\pi/2} \left(\frac{\sin x + \cos x}{\sin x + \cos x}\right) dx = \int_{0}^{\pi/2} (1)\,dx = \frac{\pi}{2} - 0$

Q. $I=\int_{\pi/6}^{\pi/3}\left(\frac{1}{1+\sqrt{\cot x}}\right)dx=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$ ——— (1)

$a=\frac{\pi}{6}, b=\frac{\pi}{3}$ & $x \to \frac{\pi}{6}+\frac{\pi}{3}-x$

or $x \to \left(\frac{\pi}{2}-x\right)$

Using prop (4): $I=\int_{\pi/6}^{\pi/3}\left(\frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\right)dx$ ——— (2)

Adding; $2I=\int_{\pi/6}^{\pi/3}(1)\,dx=(x)_{\pi/6}^{\pi/3}=\frac{\pi}{3}-\frac{\pi}{6}=\frac{\pi}{6}$

$\Rightarrow I=\frac{\pi}{12}$ Ans

Q $I=\int_1^2 \frac{\sqrt{x}}{\sqrt{3-x}+\sqrt{x}}\,dx = ?$ — (1)

$a=1,\ b=2$ so $(a+b-x)=(3-x)$

i.e. $x \longrightarrow (3-x)$ in (1)

$$I=\int_1^2 \frac{\sqrt{3-x}}{\sqrt{x}+\sqrt{3-x}}\,dx \quad \text{— (2)}$$

$$2I=\int_1^2 (1)\,dx=(x)_1^2=2-1=1$$

$$I=\frac{1}{2}$$

Q $I=\int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x}+e^{-\cos x}}\,dx$ — (1)

$x \longrightarrow (\pi-x)$

$$I=\int_0^{\pi} \frac{e^{-\cos x}}{e^{-\cos x}+e^{\cos x}}\,dx \quad \text{— (2)}$$

$$2I=\int_0^{\pi} \frac{e^{\cos x}+e^{-\cos x}}{e^{\cos x}+e^{-\cos x}}\,dx$$

$$=\int_0^{\pi}(1)\,dx=\pi \Rightarrow I=\frac{\pi}{2}$$

Quest Based on Prop 6 →

Q. $I=\int_0^{\pi} x\cos^2 x\,dx = ?$

By Prop (5), $I=\int_0^{\pi}(\pi-x)\cos^2(\pi-x)\,dx$

$=\int_0^{\pi}(\pi-x)\cos^2 x\,dx$

$I=\int_0^{\pi}\pi\cos^2 x\,dx-\int_0^{\pi}x\cos^2 x\,dx$

$2I=\pi\int_0^{\pi}\left(\frac{1+\cos 2x}{2}\right)dx$

$I=\frac{\pi}{4}\left[x+\frac{\sin 2x}{2}\right]_0^{\pi}$

$=\frac{\pi}{4}\left[(\pi-0)+(0-0)\right]=\frac{\pi^2}{4}$

Q. $I=\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}dx = ?$ (1)

By Prop (5), $x\to(\pi-x)$ i.e.

$I=\int_0^{\pi}\frac{(\pi-x)\cdot\sin x}{1+\cos^2 x}dx=\pi\int_0^{\pi}\frac{\sin x}{1+\cos^2 x}dx-I$

$2I=\pi\int_1^{-1}\frac{-dt}{1+t^2}=\pi\int_{-1}^{1}\frac{dt}{1+t^2}=\pi\left[\tan^{-1}(t)\right]_{-1}^{1}$

Put $\cos x=t$

$\sin x\,dx=-dt$

At $x=0$, $t=1$ & at $x=\pi$, $t=-1$

$2I=\pi\left(\frac{\pi}{4}-\left(-\frac{\pi}{4}\right)\right)\Rightarrow I=\frac{\pi^2}{4}$

Quest Based on Prop 7 → $\int_0^{2a} f(x)\,dx = \int_0^{a} \left(f(x)+f(2a-x)\right)dx$

Q. $I=\int_0^{2\pi} \cos^5 x\,dx = ?$

$=\int_0^{\pi}\left[\cos^5 x + \cos^5(2\pi-x)\right]dx$

$=\int_0^{\pi} \cos^5 x + (\cos^5 x)\,dx$

$=2\int_0^{\pi} \cos^5 x\,dx$

$=2\left[\int_0^{\pi/2}\left[\cos^5 x+\cos^5(\pi-x)\right]dx\right.$

$=2\int_0^{\pi/2}\left(\cos^5 x-\cos^5 x\right)dx=0$

Q. $I=\int_0^{2\pi} |\cos x|\,dx = ? = \int_0^{\pi}\left(|\cos x|+|\cos(2\pi-x)|\right)dx$

$=2\int_0^{\pi}|\cos x|\,dx = 2\int_0^{\pi/2}\left[|\cos x|+|\cos(\pi-x)|\right]dx$

$=2\int_0^{\pi/2}\left(|\cos x|+|-\cos x|\right)dx = 4\int_0^{\pi/2}|\cos x|\,dx$

$=4\int_0^{\pi/2}(+\cos x)\,dx = 4(\sin x)_0^{\pi/2} = 4$

Prop 3 e

$$\int_{-a}^{a} f(x)\,dx = \int_{0}^{a} [f(x)+f(-x)]\,dx = \begin{cases} 2\int_0^a f(x)\,dx, & \text{Even } f \\ 0, & \text{odd } f \end{cases}$$

Q. $I = \int_{-\pi/2}^{\pi/2} |\sin x|\,dx = ?$ (Even fun^n)

$$= \int_0^{\pi/2} \left(|\sin x| + |\sin(-x)|\right) dx$$

$$= \int_0^{\pi/2} \left(|\sin x| + |-\sin x|\right) dx$$

$$= 2\int_0^{\pi/2} |\sin x|\,dx = 2\int_0^{\pi/2} (+\sin x)\,dx$$

$$= 2\left(-\cos x\right)_0^{\pi/2} = -2(0-1) = 2$$

Q. $I = \int_{-\pi/4}^{\pi/4} (x^3 \sin^4 x)\,dx$ (odd.)

$$= \int_0^{\pi/4} \left[(x^3 \sin^4 x) + ((-x)^3 \sin^4(-x))\right] dx$$

$$= \int_0^{\pi/4} \left[(x^3 \sin^4 x) - (x^3 \sin^4 x)\right] dx$$

$$= 0$$

Q. $I=\int_{-\pi/2}^{\pi/2}\underbrace{(\sin|x|+\cos|x|)}_{\text{Even f.}}dx=?$

$$=\int_0^{\pi/2}\left[(\sin|x|+\cos|x|)+(\sin|-x|+\cos|-x|)\right]dx$$

$$=\int_0^{\pi/2}2(\sin|x|+\cos|x|)dx=2\int_0^{\pi/2}(\sin x+\cos x)dx$$

$$=2\left[-\cos x+\sin x\right]_0^{\pi/2}$$

$$=2\left[(0+1)-(-1+0)\right]=4$$

(B) $I=\int_{-1}^{1}\underbrace{\log\left(\frac{2-x}{2+x}\right)}_{\text{odd.}}dx=?$

$$=\int_0^1\left\{\log\left(\frac{2-x}{2+x}\right)+\log\left(\frac{2+x}{2-x}\right)\right\}dx$$

$$=\int_0^1\left[\log\left(\frac{2-x}{2+x}\right)+\log\left(\frac{2-x}{2+x}\right)^{-1}\right]dx$$

$$=\int_0^1\left[\log\left(\frac{2-x}{2+x}\right)-\log\left(\frac{2-x}{2+x}\right)\right]dx$$

$$=0$$

(9) if $f(x)$ is periodic func^n with period T then $\int_0^{nT} f(x)dx = n.\int_0^{T} f(x)dx$

Ex: $I = \int_0^{400\pi} (\sin x)\,dx = ?$ $= \int_0^{200(2\pi)} (\sin x)\,dx = 200\int_0^{2\pi} (\sin x)\,dx$

(Period of $\sin x$ is 2π)

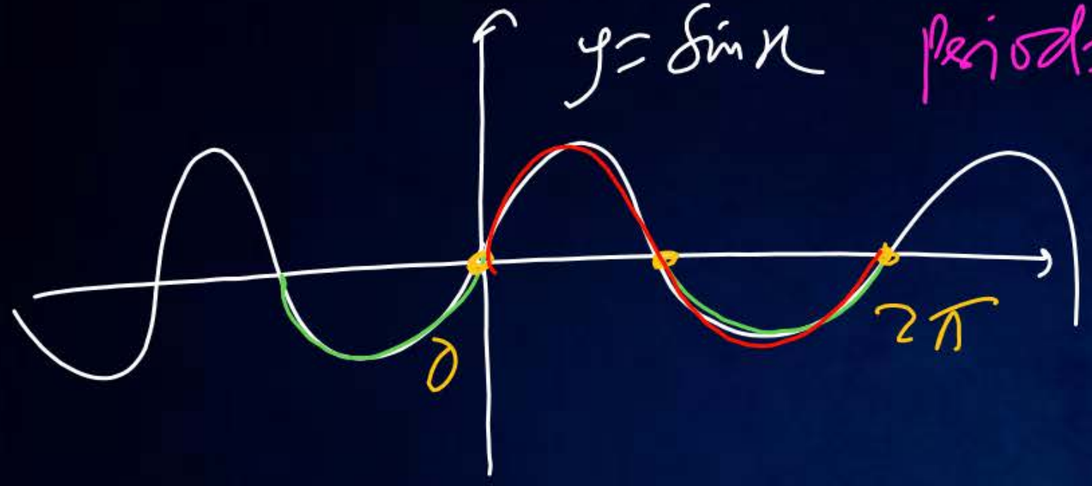
$= 200\int_0^{\pi} \{(\sin x) + \sin(2\pi - x)\}\,dx = 200\int_0^{\pi} (\sin x - \sin x)\,dx = 0$

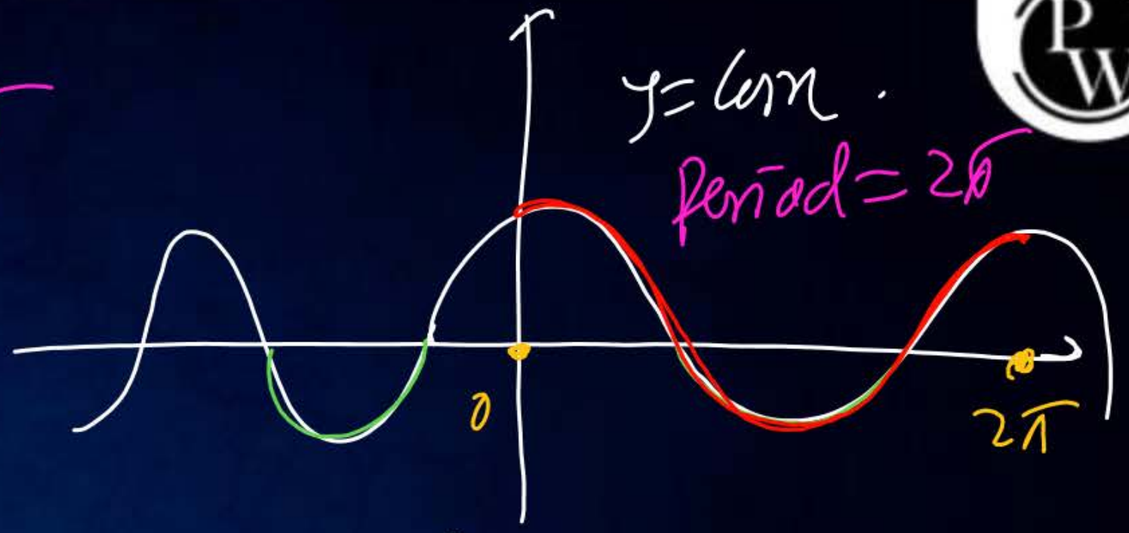
Ex: $I = \int_0^{400\pi} |\sin x|\,dx = 400\int_0^{\pi} |\sin x|\,dx = 400\int_0^{\pi/2} (|\sin x| + |\sin(\pi - x)|)\,dx$

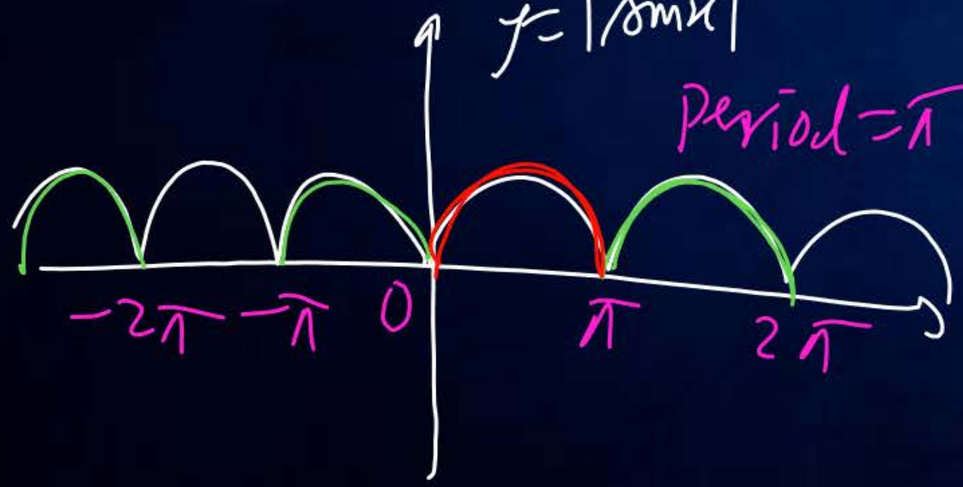
(Period of $|\sin x| = \pi$)

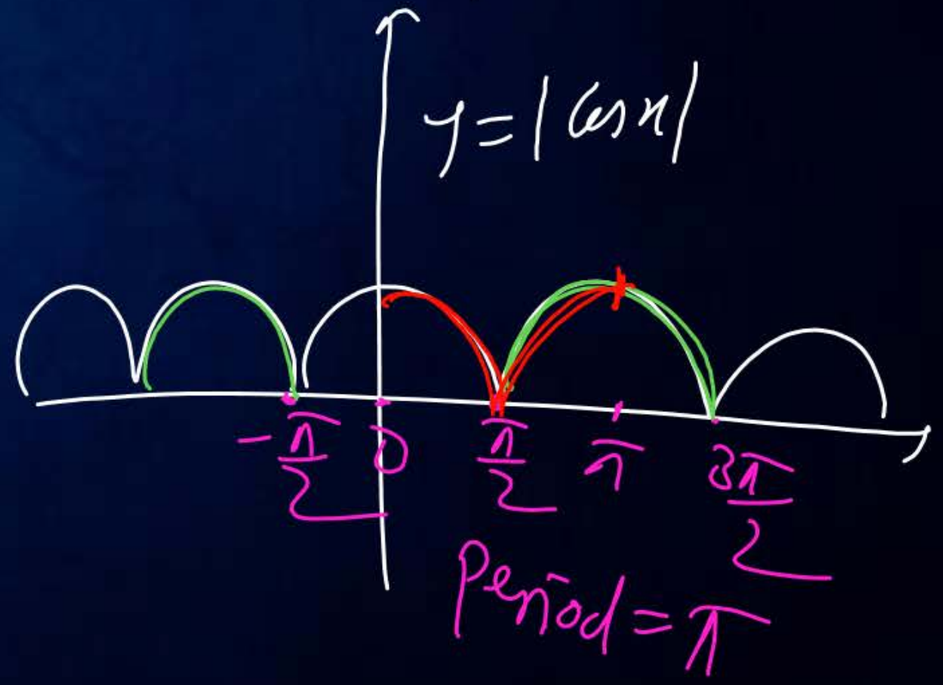
$= 400 \times 2\int_0^{\pi/2} |\sin x|\,dx = 800\int_0^{\pi/2} (+\sin x)\,dx = 800(-\cos x)_0^{\pi/2}$

$= 800.$


$y = \sin x$
Period $= 2\pi$
0
2π


$y = \cos x$.
Period $= 2\pi$
0
2π


$y = |\sin x|$
Period $= \pi$
-2π
$-\pi$
0
π
2π


$y = |\cos x|$
$-\frac{\pi}{2}$
0
$\frac{\pi}{2}$
π
$\frac{3\pi}{2}$
Period $= \pi$


Thank
YOU